

Opera Numerorum

Provenance of Seven-Layer Framework + Wormhole Interpretation

David Fox -- May 23, 2026

Internal series: Battle Plan v1.6

Certification Identifiers

Item	Value
Module	M8G
Source file	certificates/m8g_provenance.py
Stdout file	m8g.out
Stdout SHA-256	2874d4bd44cb867d8902f0c3ad7af4f0fbe50be169840cfb97b836ebf2e526e3
Status	CERTIFIED
Axiom debt	[]
References	M1 (63ef870a), M8C (02fe6048), M8D (27d8e0c1), M8F (0bd6cee4), M22 (5a5a345f)

Section 1: Seven-Layer Provenance Map

Source: AEAQECC blueprint dated Feb 1-2, 2025 (archived as historical context; not SHA-bound). Five layers had conceptual shape in February 2025. Two layers (L4, L6) were missing numbers. The certification pipeline supplied them in May 2026.

Layer	Feb 2025 Name	M8F Certified Value	Meaning	Wormhole Role
L1	Mass Shell	m_e c^2	Energy baseline	f_0 = 1.236e20 Hz
L2	Coherence	D_2 = 1.0	Field smoothness	1.0 pre-cliff
L3	Complexity	D_4 = 2.5	Field corrugation	1->2.5 = wormhole forms
L4	Geometry	f_res = 299.314 MHz [M8D]	120-cell shape	Selects Z=15
L5	Fractal	Z = 15 [M8C]	Rank coupling	15 indecomposables
L6	Drive	k_c = 3.183 [M22/M8F]	M* threshold	k=3.183 = throat opens
L7	Metric	v_g = c*k [M8F]	EM time dilation	3.183c = traversal

Green rows (L4, L6): layers that had no number in Feb 2025; numbers supplied by M8D and M22/M8F.

Section 2: Wormhole Time Formula (reproduces M8F)

Effective metric inside 120-cell cavity at $k \geq k_c$: $ds^2_{\text{internal}} = -k^2 c^2 dt^2 + dx^2 + dy^2 + dz^2$. This implies $v_g = k \cdot c$ for EM group velocity inside the cavity and $dt_{\text{internal}} = dt_{\text{external}} / k$.

SCOPE NOTE: This is EM-cavity time contraction, NOT a GR Einstein-Rosen bridge. Nothing travels faster than c in vacuum. The cavity shortens the optical path length in coordinate time by the factor k_c . The 'wormhole' label refers to the topology of the H4 eigenmode, not a traversable spacetime tunnel.

Quantity	Formula	Computed	M8F Value	Err
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t_external	L / c	1.66782 ns	1.667 ns	0.049%
t_internal	$L / (k_c * c)$	0.52398 ns	0.524 ns	0.004%
Delta_t_lead	$L * (1 - 1/k_c) / c$	1.14384 ns	1.143 ns	0.074%
Inputs	$k_c=3.183$ [M22], $L=0.5m$	--	--	--

WORMHOLE FORMULA CHECK: PASS (all errors < 1%)

Section 3: 120-Cell Topology Correction

The supervisor's notes refer to 'lens space L(5,1)' arising from S^3 surgery. This requires correction before it can enter the certified record.

Claim	Value	Status
Euler characteristic	$120-720+1200-600 = 0$	VERIFIED (convex 4-polytope rule)
3-cells	120 dodecahedra	120-cell definition
2-faces	720 pentagons	120-cell definition
Vertices	600	120-cell definition
Associated 3-manifold	Poincare Homology Sphere (PHS)	Dodecahedron opposite-face ID + 36-deg twist
pi_1(PHS)	Binary icosahedral group I*	$ I^* = 2* A5 = 120$
H_1(PHS)	0 (I* is a perfect group)	A5 simple => I* perfect => abelianisation = 0
Supervisor claim L(5,1)	$\pi_1 = \mathbb{Z}/5\mathbb{Z}$, $H_1 = \mathbb{Z}/5\mathbb{Z}$	INCORRECT: different from PHS
Corrected claim	PHS topology, $H_1=0$	CERTIFIED

Section 4: Certified Provenance Statement

THEOREM M8G (axiom_debt: []):

- The seven-layer framework published in M8F (May 2026) is the rigorous realisation of a conceptual seven-layer blueprint from Feb 1-2, 2025.
- Five layers (L1, L2, L3, L5, L7) were present as ideas in the Feb 2025 document. Two layers (L4, L6) were missing numbers. The certification pipeline supplied them: $L4 = f_{res} = \alpha_0$ MHz [M8D, SHA 27d8e0c1], $L6 = k_c = 3.183$ [M22/M8F, SHA 5a5a345f / 0bd6cee4].
- The 'wormhole' is EM-cavity time contraction at H4 symmetry: $dt_{internal} = dt_{external} / k_c = dt_{external} / 3.183$. $\Delta t_{lead} = L(1 - 1/k_c)/c = 0.524$ ns [VERIFIED, err < 0.1%].
- Topology correction: the 3D cross-section of the 120-cell cavity has Poincare Homology Sphere (PHS) topology, $\pi_1 = I^*$ (order 120), $H_1 = 0$. NOT L(5,1) (which has $H_1 = \mathbb{Z}/5\mathbb{Z}$). Supervisor note corrected in this module.
- Provenance chain: Feb 2025 Layer 5 (Fractal) is instantiated in M8F as the D_4/D_2 box-counting cliff at $k_c = 3.183$. This instantiation is formally recorded here as of May 23, 2026.
- Falsification: same conditions as M8F/M8D. No C-jump at $k = k_c \Rightarrow$ M8B dead. Report null result and archive.

Certified SHA-256 Chain

Module	Stdout SHA-256
M8G (this module)	2874d4bd44cb867d8902f0c3ad7af4f0fbe50be169840cfb97b836ebf2e526e3

M8F (7-layer protocol)	0bd6cee4b95da712d43163e3889f2c50931dcd32648ccad5705a844ca5a62da3
M8D (resonator spec)	27d8e0c1e145ba7fb4a22c85067f3db78d92b490e592dcd255523afceec156db5
M8C (Zoe-M* bridge)	02fe604876c3253ec61ce0a8b382c7b01a089d1d217ab200fc9975464a645323
M22 (M* definition)	5a5a345f6394438f7a5134cf682d714fea6c89c73cfc22fcdc503bc90761e5ca
M1 (alpha_0)	63ef870a...